# EEC 201 CA 4

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## 1 Frequency Modulated Signals

### Question 1

$$f_s = 5 \times 10^6 \text{ Hz}$$
$$T = 200\,\mu s = 200 \times 10^{-6} \text{ s.}$$
$$N = f_s T.$$

$$N = (5 \times 10^6)(200 \times 10^{-6}) = 1000.$$

Thus, the discrete-time signal contained 1000 samples.

I then expressed the time variable in discrete form. For a sampled signal,

$$t = \frac{n}{f_s}, \qquad n = 0, 1, 2, \ldots, 999.$$

$$x(t) = \cos\left(2\pi\mu t^2\right),$$

$$\mu = 4.0 \times 10^9.$$

I then substituted $t = \frac{n}{f_s}$ into the equation:

$$x[n] = \cos\left(2\pi\mu\left(\frac{n}{f_s}\right)^2\right).$$

$$x[n] = \cos\left(2\pi(4 \times 10^9)\left(\frac{n}{5 \times 10^6}\right)^2\right).$$

$$\left(\frac{n}{5 \times 10^6}\right)^2 = \frac{n^2}{25 \times 10^{12}}.$$

$$x[n] = \cos\left(2\pi(4 \times 10^9)\frac{n^2}{25 \times 10^{12}}\right).$$

$$\frac{4 \times 10^9}{25 \times 10^{12}} = 1.6 \times 10^{-4}.$$

Therefore, the discrete-time chirp signal became

$$x[n] = \cos\left(2\pi(1.6 \times 10^{-4})n^2\right), \qquad 0 \leq n \leq 999.$$

Next, I determined the instantaneous frequency of the chirp. The phase of the signal was

$$\theta(t) = 2\pi\mu t^2.$$

The instantaneous frequency is defined as

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt}.$$

$$\frac{d\theta(t)}{dt} = 4\pi\mu t.$$

$$f_i(t) = \frac{1}{2\pi}(4\pi\mu t) = 2\mu t.$$

At the start of the signal,

$$f_i(0) = 0.$$

At the end of the signal ($t = T = 200 \times 10^{-6}$),

$$f_i(T) = 2\mu T.$$

Substituting the values,

$$f_i(T) = 2(4 \times 10^9)(200 \times 10^{-6}).$$

$$f_i(T) = 1.6 \times 10^6 \text{ Hz}.$$

Thus, the chirp swept from

$$0 \text{ Hz} \quad \text{to} \quad 1.6 \text{ MHz}.$$

Finally, I verified that this frequency range did not exceed the Nyquist limit. The Nyquist frequency was

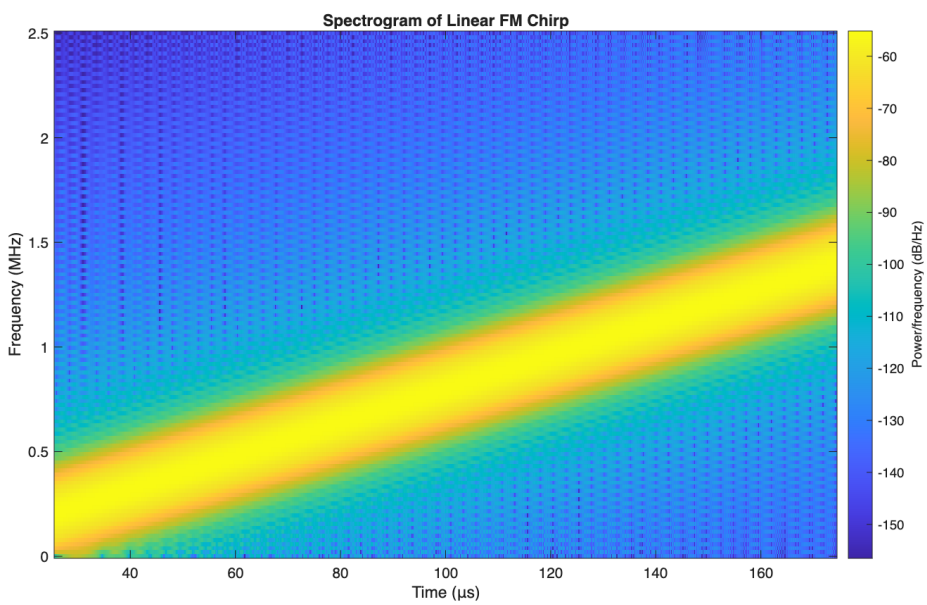
$$f_N = \frac{f_s}{2} = \frac{5 \times 10^6}{2} = 2.5 \text{ MHz}.$$

Since 1.6 MHz $<$ 2.5 MHz, the signal did not experience aliasing.

### Code

```
%% Frequency Modulated Signals Question 1)
% 1. Generate a DT linear FM chirp signal sampled at 5 MHz for 200 us
% using provided equation
% 2. Convert the CT signal to DT using t = n/fs (fs = 5 * 10^6)
% 3. Find STFT of signal using 256-point FFT, 256-sample tri window,
% 255-sample overlap
% 4. Plot frequency vs time (visualize chirp's frequency over time)

fs = 5e6;
mu = 4e9;
T = 200e-6;

N = fs*T;
n = 0:N-1;
t = n/fs;

x = cos(2*pi*mu*t.^2);

figure;
spectrogram(x, triang(256), 255, 256, fs, 'yaxis');
title('Spectrogram of Linear FM Chirp');
```

### Spectogram of the Linear FM Chip



## Question 2

In part (1), the chirp signal was given as

$$x(t) = \cos(2\pi\mu t^2),$$

where

$$\mu = 4 \times 10^9.$$

### First Definition

The first definition rewrites the signal in the form

$$x(t) = \cos(2\pi f_1(t)t).$$

$$x(t) = \cos(2\pi\mu t^2) = \cos(2\pi(\mu t)t).$$

$$x(t) = \cos(2\pi f_1(t)t),$$
$$f_1(t) = \mu t.$$

$$f_1(t) = 4 \times 10^9\, t.$$

### Second Definition

The second definition defines instantaneous frequency as the derivative of the phase. If

$$x(t) = \cos[\phi(t)],$$

then the instantaneous frequency is

$$f_2(t) = \frac{1}{2\pi}\frac{d\phi(t)}{dt}.$$

For the chirp signal,

$$\phi(t) = 2\pi\mu t^2.$$

$$\frac{d\phi(t)}{dt} = \frac{d}{dt}(2\pi\mu t^2) = 4\pi\mu t.$$

$$f_2(t) = \frac{1}{2\pi}(4\pi\mu t).$$

$$f_2(t) = 2\mu t.$$

$$f_2(t) = 8 \times 10^9\, t.$$

### Comparison of the Two Definitions

The two definitions of instantaneous frequency were therefore

$$f_1(t) = \mu t$$

and

$$f_2(t) = 2\mu t.$$

Using $T = 200\,\mu s = 200 \times 10^{-6}$:
For the first definition:

$$f_1(T) = \mu T = (4 \times 10^9)(200 \times 10^{-6}) = 8 \times 10^5.$$

$$f_1(T) = 0.8 \text{ MHz}.$$

For the second definition:

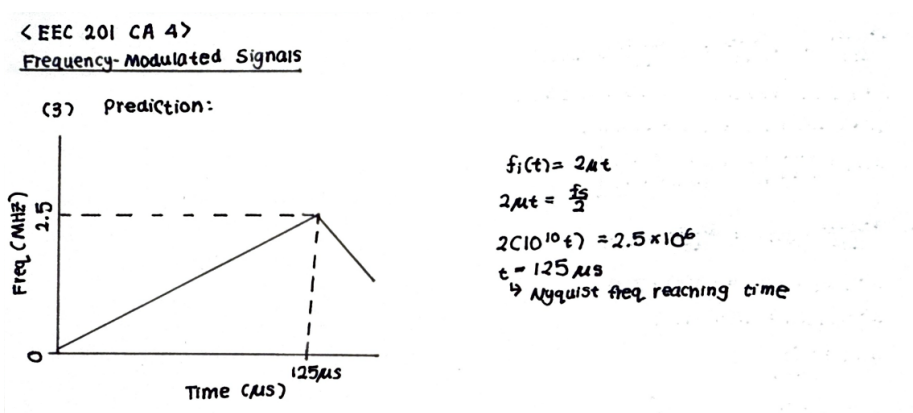
$$f_2(T) = 2\mu T = (8 \times 10^9)(200 \times 10^{-6}) = 1.6 \times 10^6.$$

$$f_2(T) = 1.6 \text{ MHz}.$$

From the spectrogram obtained in part (1), I saw that the ridge of the chirp increased from approximately 0 Hz to 1.6 MHz during the $200\,\mu s$ interval.Thus the slope of the ridge matched the second definition of instantaneous frequency, so the instantaneous frequency corresponding to the ridge in the spectrogram is given by the derivative of the phase.

## Question 3

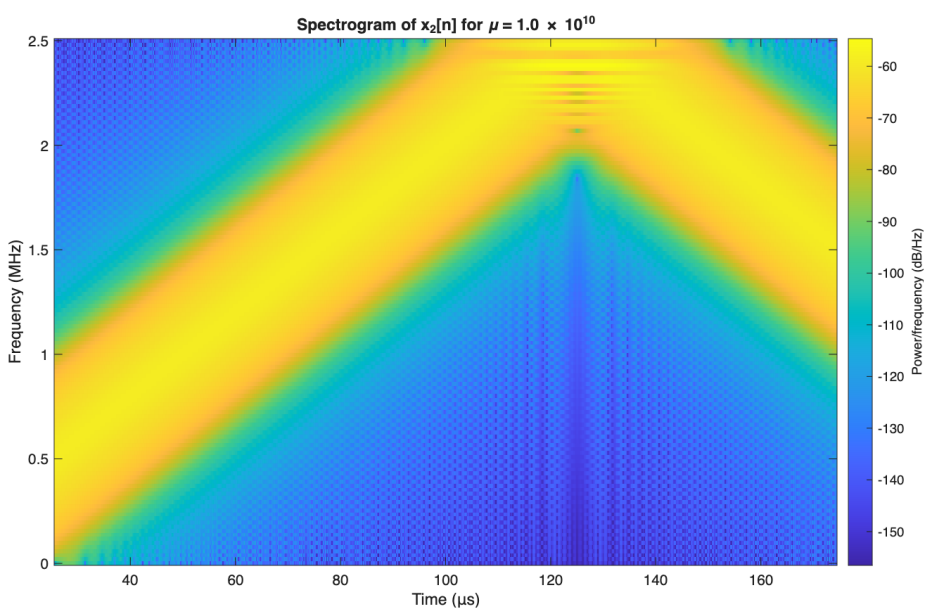
### Prediction



### Code for new chirp

```
%% Frequency Modulated Signals Question 3)
% 1. Change the chirp rate to 1 * 10^10 and predict how the spectogram
% ridge will change
% 2. Generate a new DT chirp signal using this new chirp rate
% 3. Find the spectogram of this new signal
% 4. Compare this spectogram to the one from part (1) and explain
% how the larger chirp rate changes the slope and freq of ridge.

fs = 5e6;
mu2 = 1e10;
T = 200e-6;

N = fs * T;
n = 0:N-1;
t = n / fs;

x2 = cos(2*pi*mu2*t.^2);

figure;
spectrogram(x2, triang(256), 255, 256, fs, 'yaxis');
title('Spectrogram of x_2[n] for \mu = 1.0 \times 10^{10}');
```

### Spectogram of x2[n]



**Comparison**

The first difference between the two spectograms lay in that the ridge in the new spectrogram was significantly steeper. In part (1), the chirp rate was smaller, so the dominant frequency increased more gradually over time. As a result, the ridge in the spectrogram rose smoothly and remained below the Nyquist frequency throughout the entire $200\,\mu s$ interval, reaching approximately 1.6 MHz by the end of the signal.

Secondly, the new spectrogram clearly showed aliasing (as predicted). When the chirp rate was increased to $\mu = 1.0 \times 10^{10}$, the instant frequency increased much more rapidly. Consequently, the ridge reached the Nyquist frequency of 2.5 MHz at approximately $125\,\mu s$. After this point, the ridge did not continue to increase. Instead, it folded back downward in frequency, which is the characteristic behavior that occurs when the true signal frequency exceeds the Nyquist limit and is reflected into the observable frequency range.

Thus, compared to the spectrogram from part (1), the new spectrogram initially rose more rapidly, reached the upper frequency limit sooner, and then turned downward due to aliasing. From this observation, I concluded that increasing the chirp rate makes the spectrogram ridge steeper, but if the instantaneous frequency exceeds $f_s/2$, the spectrogram no longer displays the true frequency trajectory and instead shows its aliased version.

# 2 Narrowband and Wideband Spectrograms

## Question 4

I used a relatively long analysis window so that the frequency resolution would be high. This allows the individual harmonics of the voiced speech signal to appear clearly as horizontal lines in the spectrogram. I selected the following spectrogram parameters:

$$N_w = 512, \qquad N_{\text{FFT}} = 1024, \qquad N_{\text{overlap}} = 511.$$

A triangular window was used, as specified in the problem statement, and the FFT length was chosen to be a power of two.

With these parameters, the spectrograms of both $s_1$ and $s_5$ showed clear horizontal harmonic structures. In a narrowband spectrogram, the harmonics occur at integer multiples of the fundamental frequency:

$$f = kf_0, \qquad k = 1, 2, 3, \ldots$$

Thus, the fundamental frequency can be estimated from the spacing between adjacent harmonic lines.

### Estimation of $f_0(t)$ for $s_1$

From the spectrogram of $s_1$, I observed several harmonics within the frequency range below approximately 0.5 kHz. By estimating the spacing between these harmonics, I found that the fundamental frequency was approximately

$$f_0(t) \approx 110\text{–}130 \text{ Hz}.$$

The spacing between harmonics remained relatively consistent across most segments of the signal, indicating that the pitch of the speech signal remained fairly stable over time.

### Estimation of $f_0(t)$ for $s_5$

For the signal $s_5$, the harmonic spacing in the spectrogram was noticeably larger. I estimated the fundamental frequency to be about:

$$f_0(t) \approx 200\text{–}240 \text{ Hz}.$$

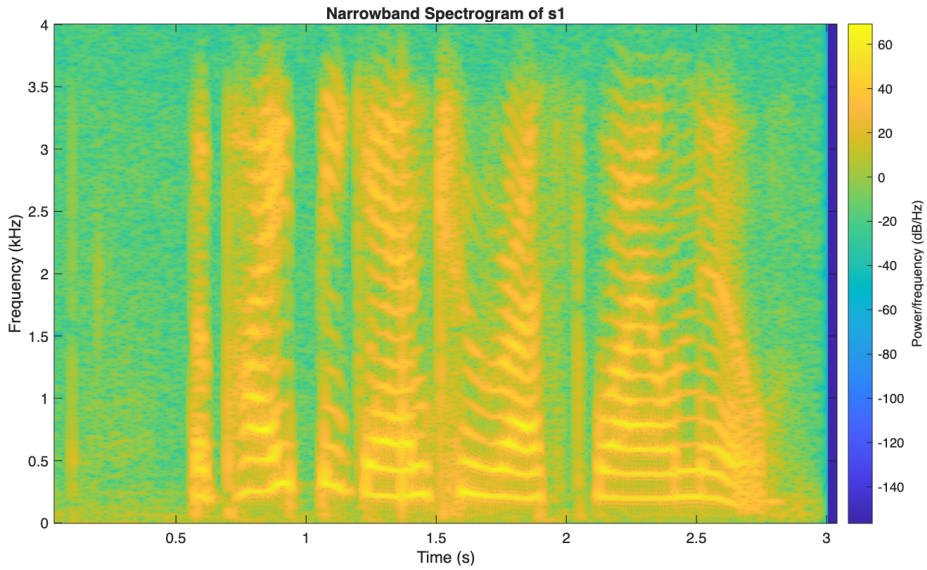
### Discussion

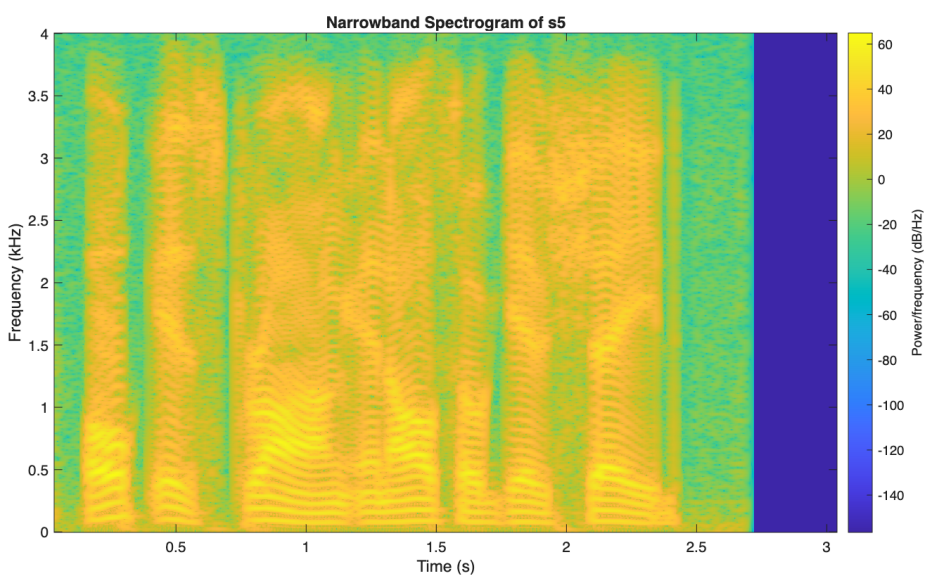
From the analysis, signal $s_1$ had a lower fundamental frequency, approximately 110–130 Hz, while $s_5$ had a higher fundamental frequency, approximately 200–240 Hz. This difference in pitch is consistent with typical speech signals, where male speakers often exhibit lower fundamental frequencies than female speakers.

### Code

```
% 1. Load the speech signals s1.mat and s5.mat (sampled at 8 kHz)
% 2. Construct narrowband spectrograms for both signals
% using a triangular window and FFT lengths that are powers of two
% 3. window length, FFT length, and overlap to obtain
% high frequency resolution (narrowband)
% 4. Use these spectrograms to estimate the fundamental
% frequency (pitch) as it changes over time.
% 5. Sketch the fundamental frequency trajectory
% for both speech signals

load s1
load s5

fs = 8000;

Nw = 512;
Nfft = 1024;
Noverlap = 511;
```

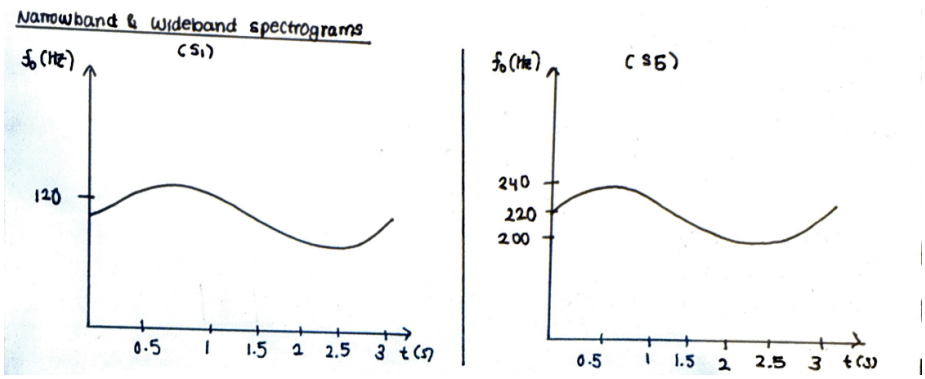
```
figure;
spectrogram(s1, triang(Nw), Noverlap, Nfft, fs, 'yaxis');
title('Narrowband Spectrogram of s1');

figure;
spectrogram(s5, triang(Nw), Noverlap, Nfft, fs, 'yaxis');
title('Narrowband Spectrogram of s5');
```



**Sketch of $f_0(t)$**

Finally, I sketched the estimated trajectories of $f_0(t)$ for both signals based on the observed harmonic spacing in the spectrograms.




## Question 5

I selected the following parameters:

$$N_w = 64, \qquad N_{\text{FFT}} = 256, \qquad N_{\text{overlap}} = 60.$$

The window length of 64 samples corresponds to a time duration of

$$\frac{64}{8000} = 8 \text{ ms},$$

which provides good time resolution for observing rapid changes in the speech signal. The FFT length was chosen to be 256, which is a power of two as required.

**Formant Estimation for $s_1$**

$$F_1(t) \approx 250\text{–}800 \text{ Hz}.$$

$$F_2(t) \approx 900\text{–}1800 \text{ Hz}.$$

$$F_3(t) \approx 2300\text{–}3200 \text{ Hz}.$$

**Formant Estimation for $s_5$**

$$F_1(t) \approx 250\text{–}700 \text{ Hz}.$$

$$F_2(t) \approx 1000\text{–}2000 \text{ Hz}.$$

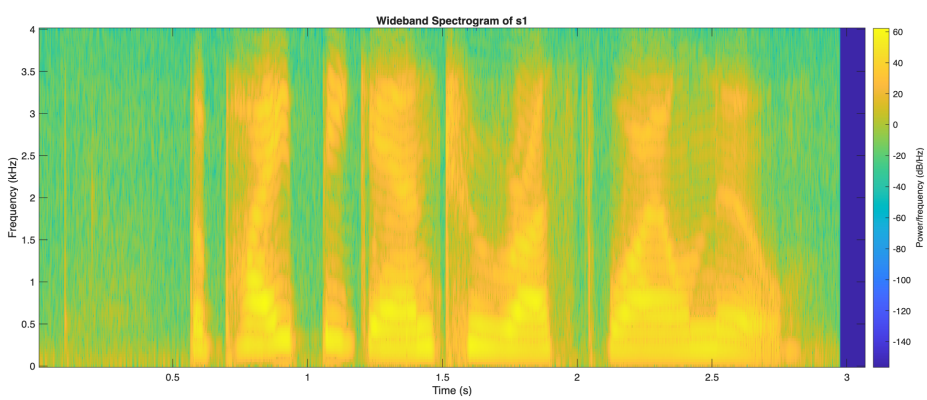
$$F_3(t) \approx 2600\text{–}3400 \text{ Hz}.$$

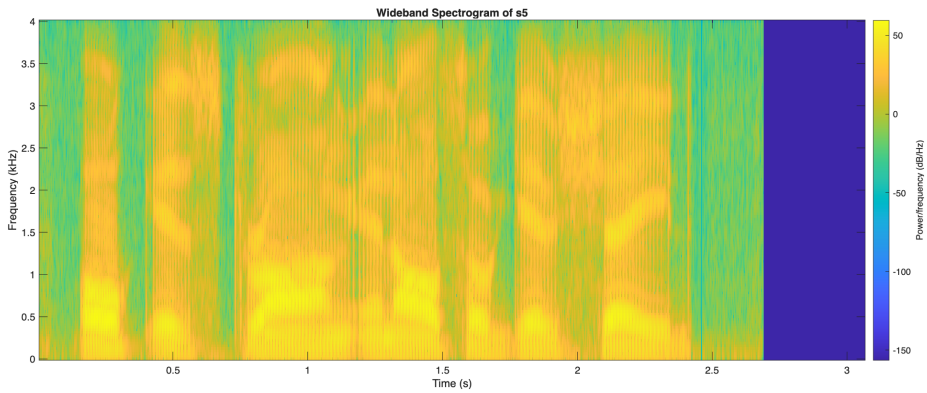
### Discussion

The wideband spectrogram provided better temporal resolution than the narrowband spectrogram used in Question 4. While the harmonic structure was no longer clearly visible, the broader spectral envelopes associated with the vocal tract resonances became easier to identify. This allowed the formant frequencies $F_1(t)$, $F_2(t)$, and $F_3(t)$ to be estimated and sketched as functions of time for both speech signals.

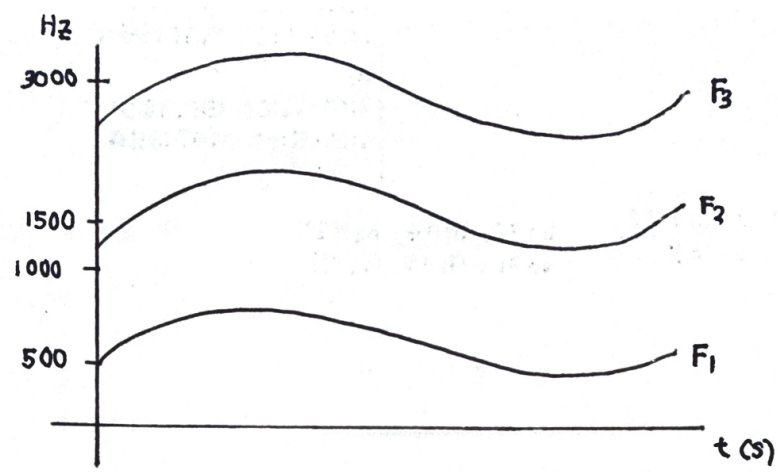
### Code

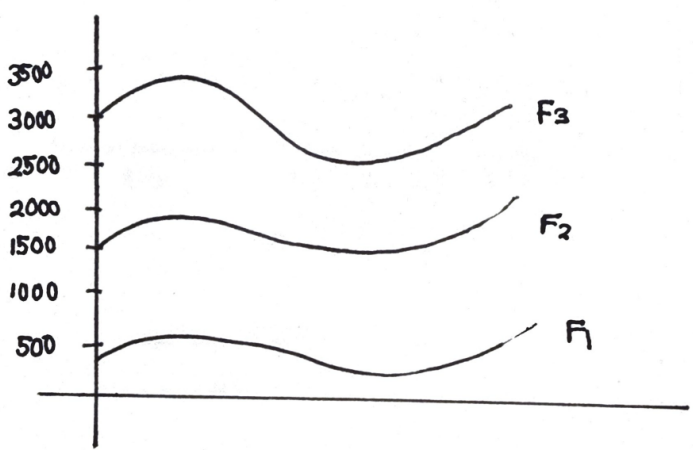
```
load s1
load s5

fs = 8000;

Nw = 64;
Nfft = 256;
Noverlap = 60;

figure;
spectrogram(s1, triang(Nw), Noverlap, Nfft, fs, 'yaxis');
title('Wideband Spectrogram of s1');

figure;
spectrogram(s5, triang(Nw), Noverlap, Nfft, fs, 'yaxis');
title('Wideband Spectrogram of s5');
```






The sketches of the formant frequencies were as follows:



# 3 Modified Short-Time Fourier Transforms

## Question 6

In general, each STFT column corresponds to a 256-sample segment of the signal, and adjacent segments are separated by a hop size of

$$R = 256 - 128 = 128$$

samples.

Also since the `spectrogram` function only returned the positive frequencies for real-valued signals, the negative-frequency components had to be reconstructed using conjugate symmetry. After reconstructing the full spectrum, I found a 1024-point inverse FFT of each column. The inverse FFT produces a 1024-sample time-domain segment, but only the first 256 samples correspond to the original windowed signal segment, so I kept only those samples.

### Code for istft_estimate.m

```
function y = istft_estimate(Xmod, L)

    Nfft = 1024;
    Nw = 256;
    R = 128;

    [K, M] = size(Xmod);

    if K ~= 513
        error('Xmod must have 513 rows for a 1024-point FFT of a real signal.');
    end

    % output accumulation arrays
    y_sum = zeros(L,1);
    y_count = zeros(L,1);

    for m = 1:M
        % positive-frequency part
        Xpos = Xmod(:,m);

        % reconstructing the full conjugate-symmetric 1024-point spectrum
        % X(1) = DC, X(513) = Nyquist
        Xfull = [Xpos; conj(Xpos(512:-1:2))];

        % inverse FFT and force real output
        xseg = real(ifft(Xfull, Nfft));
```

```
        % keeping the first 256 samples only (as per instructions)
        xseg = xseg(1:Nw);

        start_idx = (m-1)*R + 1;
        end_idx = start_idx + Nw - 1;

        if start_idx > L
            break;
        end

        if end_idx > L
            valid_len = L - start_idx + 1;
            y_sum(start_idx:L) = y_sum(start_idx:L) + xseg(1:valid_len);
            y_count(start_idx:L) = y_count(start_idx:L) + 1;
        else
            y_sum(start_idx:end_idx) = y_sum(start_idx:end_idx) + xseg;
            y_count(start_idx:end_idx) = y_count(start_idx:end_idx) + 1;
        end
    end

    y_count(y_count == 0) = 1;

    % averaging the overlapping estimates
    y = y_sum ./ y_count;
end
```

To test this function, I found the spectrogram of `vowels.mat` using the same parameters and then used the STFT as input to my reconstruction function to estimate the time-domain signal. Finally, I plotted the original signal, the reconstructed signal, and the difference between them just to check.

### Code for q6_test.m

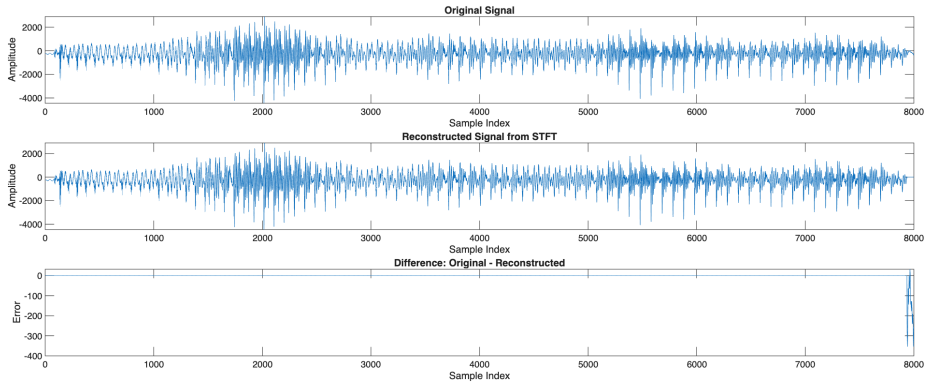
```
load vowels

x = vowels;              % original signal
L = length(x);

Nw = 256;
Nfft = 1024;
Noverlap = 128;

[S, F, T] = spectrogram(x, rectwin(Nw), Noverlap, Nfft);

% reconstruct the signal from unmodified STFT
y = istft_estimate(S, L);
```

```
% error signal
e = x - y;

figure;
subplot(3,1,1);
plot(x);
title('Original Signal');
xlabel('Sample Index');
ylabel('Amplitude');

subplot(3,1,2);
plot(y);
title('Reconstructed Signal from STFT');
xlabel('Sample Index');
ylabel('Amplitude');

subplot(3,1,3);
plot(e);
title('Difference: Original - Reconstructed');
xlabel('Sample Index');
ylabel('Error');

% numerical error check just to be safe!!
fprintf('Maximum absolute error = %g\n', max(abs(e)));
fprintf('Mean squared error = %g\n', mean(e.^2));
```

The reconstructed signal closely matched the original signal across almost the entire duration. The difference signal was nearly zero everywhere except near the end of the signal. I think this small error occurs because the final STFT window extends beyond the end of the signal and MATLAB pads the remaining samples with zeros when computing the FFT. During reconstruction, these padded samples introduce a small discrepancy near the boundary.
The numerical error values were

$$\text{Maximum absolute error} = 352$$

$$\text{Mean squared error} = 286.624$$


Original Signal
Amplitude
2000
0
-2000
-4000
0
1000
2000
3000
4000
5000
6000
7000
8000
Sample Index
Reconstructed Signal from STFT
Amplitude
2000
0
-2000
-4000
0
1000
2000
3000
4000
5000
6000
7000
8000
Sample Index
Difference: Original - Reconstructed
Error
0
-100
-200
-300
-400
0
1000
2000
3000
4000
5000
6000
7000
8000
Sample Index